Evading the Binary Tax: High-Throughput Software Engineering Micro-Agents via 1.58-Bit Co-Designed Discrete Text Diffusion with Vision: An Open-Source **Ternary Deployment Toolset**

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Abstract

Autonomous software engineering (SWE) micro-agents running natively on local edge nodes are heavily bottlenecked by the memory footprints and processing demands of traditional floating-point models—a constraint we define as the **"Binary Tax."** This paper introduces **TernaryDiffusionGemma 2.6M 1.0 (Tri-Hole Firmware)**, an open-source hardware-software co-designed computing engine engineered using the structural architecture of the Google Gemma family. Rather than relying on standard autoregressive generation, our system runs via an ultra-lightweight, 4-block bidirectional text-diffusion canvas. To ensure deep compatibility and optimization stability, our framework directly utilizes the official Gemma 256k vocabulary schema, RMSNorm normalizations, and the SwiGLU activation gating mechanism. All continuous weights are locked to a discrete ternary radix spectrum {−1, 0, +1} using Quantization-Aware Training (QAT) with a Straight-Through Estimator (STE).

To enable complete community reproduction and verification, we introduce the full-stack **Ternary Development Kit (TDK)**, which delivers our complete public repository configuration across seven integrated software, firmware, and protocol utilities:

- **`run_gemma_automaton.py`:** The native PyTorch training framework orchestrating multiplier-free canvas diffusion using an active gradient-passing interface.
- **`trit_packer.py`:** A highly space-efficient 5-to-1 trit-packing utility that compresses model memory space by ~95% down to a microscopic ~520 KB, allowing the baseline model brain (`nano_weights.bin`) to sit entirely inside hardware L3 caches.
- **`unpacker.cpp`:** A bare-metal, maximum-optimization C++ execution bridge that replaces cycle-heavy Multiply-Accumulate (MAC) routines with single-cycle conditional branch networks linked directly via static mapping configurations (`weights.h`).
- **`vision_transformer_color.py`:** A cross-modal frontend vision pipeline processing Retinex log-reflectance algorithms to extract illumination-invariant structural signatures mapped natively to 128-Trit Matryoshka Semantic Embeddings.
- **`T_plus_plus_spec.md`:** The data language serialization dictionary protocol enforcing a strict, fixed-width 6-Trit Symbol Envelope across standard ASCII formats, leaving exactly 602 unallocated custom macro states open for localized hardware-level safety lines.

Empirical benchmarks on an AMD/NVIDIA RTX 3090 desktop environment demonstrate absolute learning stability, with cross-entropy loss decaying smoothly to 5.3468 at a steady state of 145.1 Watts with zero thermal drift. When evaluated via pure software proxy emulation on a resource-constrained ARMv8-A architecture (such as a 16GB Raspberry Pi edge node), the system operates at a precision latency profile of 20.293 µs per loop. Our framework serves as a modular, user-adaptable frontend utility agent, providing zero-cost structural isolation against classic binary register rollovers while seamlessly outputting feature representations optimized for downstream multi-tier MLM (DiffusionGemma) and MRL (EmbeddingGemma) architectures at the decentralized edge.